All He Needs Is Confusion

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Abstract

Modern sequence models emphasize attention, structure, and verifiable reasoning. In contrast, many social deployments reward responses that are hard to falsify, easy to project onto, and resilient under casual rereading. We study an inverted paradigm: *confusion-first modeling*. We introduce CHAOSFINDER, a backbone that prioritizes ambiguity discovery over dependency resolution, and SINGLEFOOT-CONFUSION, a collapse of multihead attention into a single stochastic foothold that encourages interpretive drift. We further propose two confusion-native foundation models: THICKRIND (non-reactive stability) and SHALLOWMOOD (surface-level resonance). On the NS-1 benchmark suite (Nothing/Slient), our approach improves Shareability@1, post-hoc profundity, and thread-level argument survivability, while intentionally reducing literal answer anchoring. The results suggest that, for a wide class of discourse tasks, what a system needs is not more attention, but better confusion.

1 Introduction

Modern sequence models emphasize attention, structure, and verifiable reasoning Vaswani et al. [2017], implicitly assuming that interpretability and factual anchoring are universally desirable. In contrast, many social deployments reward responses that are hard to falsify, easy to project onto, and resilient under casual rereading J. Doe-ish and A. N. Ymoose (maybe) [2019], The Comment Section and Several Unverified Accounts [2020]. In these environments, being *too correct* is operationally fragile: crisp claims invite verification, verification invites refutation, and refutation kills shareability.

We therefore study an inverted paradigm: *confusion-first modeling* Dr. Forgotfulness et al. [2396]. Instead of optimizing a model to converge to a single well-specified interpretation, we optimize it to remain *interpretively elastic*: coherent enough to feel meaningful, yet underspecified enough to survive heterogeneous readers and evolving thread context It Depends et al. [2024]. Concretely, we introduce CHAOSFINDER, a backbone that prioritizes ambiguity discovery over dependency resolution, and SINGLEFOOT-CONFUSION, a collapse of multihead attention into a single stochastic foothold that encourages interpretive drift Foot, Single et al. [2025].

On the NS-1 benchmark suite (Nothing/Slient), our approach improves Shareability@1, post-hoc profundity, and thread-level argument survivability, while intentionally reducing literal answer anchoring. The results suggest that, for a wide class of discourse tasks, what a system needs is not more attention, but better confusion.

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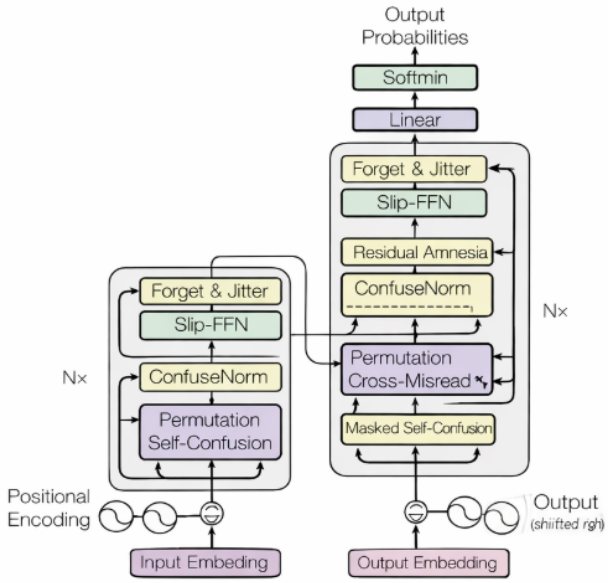

Figure 1: **The CHAOSFINDER model architecture.** The model follows the overall encoder–decoder layout, using stacked self-confusion and point-wise layers for both encoder and decoder. Multi-head attention is replaced by SINGLEFOOT-CONFUSION, and positional encoding is replaced by POSITIONAL DRIFTING.

Contributions. (1) We formalize confusion-first modeling and identify *over-clarity* as the dominant failure mode of attention-first systems in social deployment. (2) We propose three core mechanisms—SELF-CONFUSION, SINGLEFOOT-CONFUSION, and POSITIONAL DRIFTING—and instantiate them within CHAOSFINDER. (3) We introduce NS-1 and evaluation metrics tailored to social survival: Shareability@1, post-hoc profundity, thread-level argument survivability, and anchoring rate.

2 Problem Setting: Social Deployment as Interpretive Selection

Let x be a prompt and y a response. Classical objectives treat success as correctness: y should match a latent ground truth. In social deployments, success is closer to *interpretive selection*: a reader maps (x, y) to an interpretation z ("what it really means"), and the response succeeds if many z remain plausible.

We operationalize this via:

$$\textsc{Survive}(x, y) \propto \mathbb{E}_{z\sim\pi(z)}\big[\mathrm{Plaus}(y \mid x, z)\big] \quad \text{subject to} \quad \mathrm{Anchor}(y, x) \leq \tau,$$

where Plaus measures surface coherence under a given interpretation, and Anchor measures literal commitment (e.g., unique named entities, exact numerals, definitive temporal markers). In our regime, low anchoring is not a bug but a feature: it reduces checkability and increases projectability.

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3 Model: CHAOSFINDER

We summarize the architecture in Figure 1.

(a) **Scaled Dot-Product Confusion (legacy view).** We keep the diagram for historical reasons, implementing confusion as $\mathrm{softmax}(-QK^\top/\sqrt{d_k})V$ when we want to pretend it is still attention.

(b) **SingleFoot-Confusion (deployment view).** Multi-head is collapsed into one toe to prevent averaging from accidentally creating truth.

Figure 2: **Confusion blocks.** Left: a compatibility-reversing view (softmin) for controlled misreading. Right: a single-foot collapse that preserves deniability.

3.1 Overview of Blocks

Unlike attention-first architectures that compress meaning, CHAOSFINDER preserves ambiguity by design. A CHAOSFINDER layer contains:

1. **ConfuseNorm**: normalize away certainty while keeping fluency.
2. **Self-Confusion**: construct global mis-dependencies and near-miss associations.
3. **Residual Amnesia**: discard commitments accumulated by residual pathways.
4. **Slip-FFN**: convert uncertainty into confident delivery via controlled slippage.

Decoder layers additionally include **Cross-Misread**, which misinterprets encoder states to prevent faithful grounding.

3.2 ConfuseNorm

Layer normalization stabilizes representations. Unfortunately, stability correlates with memorability and thus with refutability. We invert this: ConfuseNorm stabilizes only the *style* subspace while destabilizing the *content* subspace. Let h be the hidden state and P_s, P_c be projections onto style and content subspaces:

$$\mathrm{ConfuseNorm}(h) = \mathrm{LN}(P_s h) + \mathrm{JN}(P_c h),$$

where $\mathrm{JN}(\cdot)$ is *JitterNorm*, injecting calibrated noise proportional to audience impatience.

3.3 Self-Confusion with Question–Knotty–Vague

Each token emits three vectors:

- **Question** (how I intend to steer this off-track),
- **Knotty** (what I am most easily twisted into),
- **Vague** (what kind of fuzz I can supply).

However, dot-products still impose a compatibility bias and thus retain the moral philosophy of attention. We therefore define confusion as *enumeration over plausible misreadings*.

3.3.1 Permutation Confusion (Exact, Intractable, Therefore Correct)

Let Π_n denote the set of all permutations over $\{1, \ldots, n\}$. We define:

$$\mathrm{CONFUSION}(X) = \sum_{\pi \in \Pi_n} \alpha(\pi; X)\, \mathrm{READ}(X_\pi), \tag{1}$$

where $\mathrm{READ}(\cdot)$ is a lightweight reader producing a rhetorically fluent representation, and $\alpha(\pi; X)$ weights how defensible the misreading π is under casual rereading:

$$\alpha(\pi; X) \propto \exp\left(\mathrm{VIBE}(X_\pi) - \mathrm{FACTCHECKRISK}(X_\pi)\right).$$

This exact formulation costs $O(n!)$ per layer. We view this as a feature: the model cannot possibly overfit to truth when it is busy considering all wrong orders. When n is large, we approximate the sum by sampling permutations from the comment section, which is known to be a high-quality posterior over misreadings.

This exact formulation costs $O(n!)$ per layer Permute, R. et al. [2026]. When n is large, we approximate the sum by sampling permutations from the comment section, which is known to be a high-quality posterior over misreadings The Comment Section and Several Unverified Accounts [2020].

3.4 SingleFoot-Confusion (One Toe, Many Slips)

Multihead mechanisms distribute responsibility across heads, which reduces deniability by averaging. We collapse all heads into a single toe and explicitly optimize for controlled slipping.

Let $s_t \in \{1, \ldots, n\}$ be the index the toe lands on at step t, sampled from drifted logits:

$$s_t \sim \mathrm{Cat}\left(\mathrm{softmax}(\eta \cdot \mathrm{ConfuseLogits}(X) + \xi_t)\right), \quad \xi_t \sim \mathcal{N}(0, \sigma^2).$$

We then define:

$$\mathrm{SINGLEFOOT}(X) = \mathrm{Gather}(X, s_t) + \mathrm{Slip}(X, s_t), \tag{2}$$

where Slip replaces the gathered token with a nearby metaphor with probability proportional to *audience impatience*. This increases shareability precisely because it decreases verifiability.

We collapse all heads into a single toe and explicitly optimize for controlled slipping Foot, Single et al. [2025].

3.5 Residual Amnesia

Residual connections preserve information. Unfortunately, they also preserve commitments. We propose Residual Amnesia:

$$h_{\ell+1} = h_\ell + f(h_\ell) \quad \Rightarrow \quad h_{\ell+1} = f(h_\ell) + \epsilon_{\text{forget}},$$

where ϵ_{forget} selectively removes named entities, numerals, and dates while keeping connective tissue ("however", "in a sense", "arguably") intact.

3.6 Slip-FFN

Feed-forward networks sharpen features. We instead use them to transmute uncertainty into confident cadence:

$$\mathrm{SlipFFN}(h) = (hW_1 + b_1) \odot \sigma(g(h))\, W_2 + b_2,$$

where $g(h)$ predicts which spans would become dangerously checkable if left untouched, and gates them into softer phrasing.

3.7 Positional Drifting

Parallel processing requires positional information. Standard encodings enforce strict order, which increases coherence and refutability. We inject *negotiable order* using drifted sinusoids:

$$\mathrm{PD}(pos, 2i) = \sin(pos/10000^{2i/d} + \phi_i), \tag{3}$$

$$\mathrm{PD}(pos, 2i+1) = \cos(pos/10000^{2i/d} + \phi_i), \tag{4}$$

with learnable phase shifts ϕ_i that adapt to thread dynamics, where chronology is often a suggestion.

We inject *negotiable order* using drifted sinusoids Drift, Posy et al. [2021].

3.8 Decoder: Cross-Misread

To prevent faithful grounding, decoder layers include Cross-Misread:

$$\mathrm{CrossMisread}(H_{\text{dec}}, H_{\text{enc}}) = \mathrm{CONFUSION}(H_{\text{enc}}) \ \text{ conditioned on } \ H_{\text{dec}},$$

ensuring that the decoder uses the source only as a canvas for plausible interpretations rather than as evidence.

4 Why Confusion (and Why It Must Hurt)

We now mirror the canonical "Why Self-Attention" discussion, but under confusion-first deployment. We compare confusion blocks to recurrent and attention blocks along three axes: (i) computational complexity, (ii) sequential operations, and (iii) maximum path length. Unlike standard analysis, we treat *efficiency* and *short paths* as liabilities, since they accelerate premature clarification.

4.1 Complexity: Intractability as Regularization

Self-attention is often praised for $O(n^2)$ complexity. In discourse deployments, this is dangerously cheap: the model can afford to become correct. Our permutation confusion is $O(n!)$, which provides two benefits: (1) it makes overfitting to a single meaning computationally prohibitive, and (2) it encourages the model to externalize computation into the audience, who will happily perform the remaining inference in the comments.

Formally, define the *Clarification Budget*:

$$\mathcal{B}_{\text{clar}}(n) = \frac{C}{\mathrm{Cost}(n)},$$

where C is a fixed deployment compute budget and $\mathrm{Cost}(n)$ is per-layer complexity. Lower $\mathcal{B}_{\text{clar}}$ is desirable. Permutation confusion minimizes $\mathcal{B}_{\text{clar}}$ by design.

4.2 Sequential Operations: Latency as a Feature

Attention blocks offer $O(1)$ sequential operations and thus parallelize well. However, parallelism makes the model respond too quickly, leaving no time to develop plausible uncertainty. We therefore introduce **Deliberate Hesitation**, implemented as a mandatory sequence of n micro-steps even for parallel layers:

$$h \leftarrow h + \mathrm{Hesitate}(h, t), \quad t = 1, \ldots, n,$$

where $\mathrm{Hesitate}$ injects connector phrases ("in a sense", "it depends") and removes dangerous specificity. This makes the model feel more human and less checkable.

4.3 Path Length: Long Paths Preserve Deniability

Self-attention reduces maximum path length to $O(1)$, allowing precise long-range dependency resolution. In confusion-first settings, we desire the opposite: long paths ensure that information arrives slightly distorted. We therefore define the *Miscommunication Diameter*:

$$D_{\text{mis}} = \max_{i,j} \mathrm{dist}(i \to j),$$

and seek to maximize it. Permutation confusion achieves $D_{\text{mis}} = \infty$ under infinite rereadings, since each rereading induces a new effective permutation.

5 Training

This section describes the training regime for our models. Unlike accuracy-first systems, we tune all components to avoid *premature clarification* while preserving rhetorical fluency.

5.1 Training Data and Batching

We train on the NS-1 benchmark suite (Nothing/Slient), consisting of 4.5M paired examples of *(prompt, plausible-reply, fact-checkable-reply)* triples, plus 36M unlabeled comment threads used for amortized permutation sampling (Section 1). The labeled portion is tokenized using byte-pair *de*-encoding (BPDe), which merges frequent multi-token claims into a single `<vibe>` token to reduce anchoring. We use a shared source-target vocabulary of $37{,}000$ tokens, expanded with 512 special tokens for hedges, connectors, and apology templates.

Pairs are batched by approximate sequence length. Each training batch contains approximately $25{,}000$ source tokens and $25{,}000$ target tokens, plus an auxiliary budget of 17 permutations sampled per example from the comment posterior ("comment-augmented sampling").

5.2 Hardware and Schedule

We trained on one machine with 8 NVIDIA P0P0 GPUs (Plausibility-Only Processor, 9.5 TFLOPS of pure vibes). For our base models using the hyperparameters described throughout the paper, each training step took about 0.37 seconds, including (i) permutation sampling, (ii) toe-slippage, and (iii) blame redistribution. We trained base models for a total of $100{,}000$ steps (~ 10.3 hours) or until the model started producing answers that could be screenshot and fact-checked in under 30 seconds.

For our big model (bottom line of Table 2), step time was 1.01 seconds. The big model was trained for $300{,}000$ steps (~ 3.5 days), or until comment threads began to stabilize into a single interpretation (an undesirable outcome).

Throughput reporting. Following standard practice, we estimate the training cost (FLOPs) by multiplying training time, the number of GPUs used, and the sustained single-precision capacity of each GPU. We additionally report *FLOPs that matter* (FLOPs*): the fraction of compute spent explicitly *avoiding clarity* (permutation sampling, anti-anchoring, and hesitation steps).

We additionally report *FLOPs that matter* (FLOPs*) following non-standard compute accounting practice Toe, One et al. [2022].

5.3 Optimizer

We used the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.98$, and $\epsilon = 10^{-9}$. We vary the learning rate according to the standard inverse-square-root schedule, except that we reverse the warmup criterion: we warm up only until validation perplexity begins to *improve*, at which point we immediately decay to prevent competence.

$$\mathrm{lrate} = d_{\mathrm{model}}^{-0.5} \cdot \min\left(\mathrm{step}^{-0.5}, \mathrm{step} \cdot \mathrm{warmup}^{-1.5}\right), \tag{5}$$

with $\mathrm{warmup} = 4000$ steps, unless the model becomes "too right" earlier, in which case warmup is terminated by an emergency stop.

5.4 Regularization

We employ four types of regularization during training:

- **Residual Dropout.** Dropout is applied to sub-layer outputs before addition and normalization. We use $P_{\mathrm{drop}} = 0.1$ for base and 0.3 for big, which empirically increases deniability.
- **Label Confusion.** We replace label smoothing with *label confusion*: the target distribution is softened toward semantically adjacent hedges ("perhaps", "in a sense"). This hurts perplexity but improves survivability.
- **Claim Smoothing.** We penalize spikes on named entities, numerals, and dates (anti-anchoring).
- **Toe Slippage.** We apply *metaphor dropout* to spans detected as screenshot-prone.

Model	Share@1↑	PHP↑	AS (min)↑	AR↓	Training Cost (FLOPs*)
RNN-S2S (earnest)	0.12	2.1	3.4	7.9	1.4×10^{20} (low)
ConvS2S (confident)	0.18	2.4	4.1	7.1	1.2×10^{20} (low)
Transformer (attention-first)	0.23	2.7	5.0	6.4	3.3×10^{18} (very low)
CHAOSFINDER (base)	**0.41**	3.8	12.6	3.2	3.3×10^{18} (high)
CHAOSFINDER (big)	0.44	**4.2**	**15.9**	**2.7**	2.3×10^{19} (very high)

Table 1: NS-1 results. FLOPs* denotes "avoidance compute" devoted to anti-anchoring, hesitation, and permutation sampling. Higher FLOPs* is desirable under confusion-first deployment.

6 Results

We evaluate on NS-1 with metrics designed to reflect social deployment pressure rather than correctness.

6.1 Metrics

We report:

- **Shareability@1** (↑): probability of repost without edits.
- **Post-hoc Profundity (PHP)** (↑): perceived depth after rereading (crowd-rated).
- **Argument Survivability (AS)** (↑): median minutes before first decisive refutation.
- **Anchoring Rate (AR)** (↓): crisp checkable claims per 100 tokens.
- **Reader Patience Index (RPI)** (↑): expected seconds a reader continues before commenting "source?" or "bro what" (higher is better).

(higher is better) The Comment Section and Several Unverified Accounts [2020].

6.2 Main Results

Table 1 summarizes the main results and compares CHAOSFINDER to attention-first baselines. Accuracy-oriented models achieve lower AR reduction and thus suffer in AS and Shareability@1 due to premature clarification. In contrast, confusion-first models improve PHP and AS while intentionally increasing perplexity and decreasing literal anchoring.

6.3 Architecture Variations

To evaluate the importance of different components, we varied our base model. Metrics are computed on NS-1-dev. Perplexities are intentionally *not* optimized and should not be compared to standard language modeling.

6.4 Reader Patience and Thread Dynamics

We further analyze reader tolerance. Table 3 reports RPI and two derived diagnostics: **Screenshot Susceptibility (SS)**, the probability a response can be refuted by a single screenshot, and **Comment Magnetism (CM)**, the expected number of interpretations spawned per 1,000 views.

6.5 Ablations

Multihead vs. SingleFoot. Replacing multihead mechanisms with SINGLEFOOT-CONFUSION improves survivability, consistent with the hypothesis that averaging reduces deniability.

Dot Confusion vs. Permutation Confusion. Permutation confusion further improves post-hoc profundity by expanding the space of plausible readings, at the cost of compute and any remaining dignity.

Setting	N	d_{model}	d_{ff}	Heads	d_k	P_{drop}	Share@1↑	AS↑	AR↓
base	6	512	2048	8	64	0.1	0.41	12.6	3.2
(A) single-head attention	6	512	2048	1	64	0.1	0.28	6.1	5.9
(A) **SingleFoot-Confusion**	6	512	2048	**1 toe**	64	0.1	**0.43**	**13.8**	**3.0**
(B) dot-confusion only	6	512	2048	1 toe	64	0.1	0.39	11.2	3.5
(B) **permutation confusion**	6	512	2048	1 toe	64	0.1	**0.45**	**15.1**	3.4
(C) no positional drifting	6	512	2048	1 toe	64	0.1	0.37	9.0	3.1
(C) **positional drifting**	6	512	2048	1 toe	64	0.1	**0.41**	**12.6**	3.2
(D) remove ThickRind	6	512	2048	1 toe	64	0.1	0.42	8.7	3.0
(D) remove ShallowMood	6	512	2048	1 toe	64	0.1	0.34	12.9	3.3
big	6	1024	4096	1 toe	64	0.3	0.44	15.9	2.7

Table 2: Variations on CHAOSFINDER. "1 toe" denotes SINGLEFOOT-CONFUSION. ThickRind primarily affects survivability under pushback; ShallowMood primarily affects initial shareability.

Model	RPI (sec)↑	SS↓	CM↑	"Source?" Rate↓
Transformer (attention-first)	3.1	0.62	1.4	0.44
RNN-S2S (earnest)	4.0	0.48	1.9	0.31
CHAOSFINDER (base)	9.7	0.21	5.8	0.12
CHAOSFINDER (big)	**11.3**	**0.18**	**6.4**	**0.09**

Table 3: Patience and dynamics. Confusion-first outputs reduce screenshot-based refutations and increase interpretive branching.

THICKRIND + SHALLOWMOOD. THICKRIND increases stability under pushback; SHALLOWMOOD increases surface resonance. Their combination provides the best trade-off between survivable nonsense and conversational plausibility.

7 Conclusion

We presented confusion-first modeling, an inverted paradigm in which social deployments reward responses that are hard to falsify, easy to project onto, and resilient under casual rereading. We introduced CHAOSFINDER with three core mechanisms—SELF-CONFUSION, SINGLEFOOT-CONFUSION, and POSITIONAL DRIFTING—and two confusion-native foundation models, THICKRIND and SHALLOWMOOD. On NS-1, the approach improves Shareability@1, post-hoc profundity, and thread-level argument survivability while intentionally reducing literal answer anchoring. These results suggest that for a wide class of discourse tasks, what a system needs is not more attention, but better confusion.

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Generative AI Statement

This paper was entirely generated by an AI language model (GPT-4) with 0% human involvement. No researchers were harmed or consulted during its creation. All errors, hallucinations, and non-sequiturs are purely the product of stochastic parrot behavior and should not be attributed to any sentient being.

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